PRACTICE EXAM 3 · EXAM STRATEGY - ALL DOMAINS

Practice exam 3

The last twenty questions, and a read of where this exam is heading.

TYPE Practice exam QUESTIONS 20 TOPIC All sections

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> A streaming table has ingested from a Kafka topic for eighteen months and holds 4 billion rows. The topic retains seven days. A schema change requires the pipeline logic to be rewritten, and an engineer plans a full refresh afterwards.
>
> The team asks what they should prepare for.

What is the risk?

A The full refresh will take several hours but complete correctly.

B A full refresh clears the streaming state and reprocesses from source. Everything older than seven days cannot be reprocessed and is lost.

C The full refresh will fail because the checkpoint is incompatible with the new logic.

D Nothing; streaming tables retain their own copy of source data.

◆ PAUSE AND DECIDE

What is the risk?

A The full refresh will take several hours but complete correctly.

It will complete, and with only seven days of data. The duration is not the problem.

B A full refresh clears the streaming state and reprocesses from source. Everything older than seven days cannot be reprocessed and is lost.

Retention arithmetic: eighteen months of history behind a seven-day source. The difference is unrecoverable.

C The full refresh will fail because the checkpoint is incompatible with the new logic.

A full refresh clears the checkpoint deliberately. That is what it does.

D Nothing; streaming tables retain their own copy of source data.

They hold processed output, which a full refresh discards before reprocessing.

ON THE EXAM

A full refresh that succeeds is the dangerous outcome here. It completes cleanly and the table is 4 billion rows lighter.

> A serverless wheel task fails at import time on a library that works fine in a developer's notebook. The developer installed it in the notebook's environment panel; the task uses an environment defined in the bundle.
>
> Both specify the same library.

What is the most likely cause?

A The environment defined in the bundle does not include a transitive dependency the notebook panel resolved automatically.

B Wheel tasks cannot use environments; only notebook tasks can.

C Serverless does not support third-party libraries.

D The bundle must install libraries at cluster scope rather than in an environment.

◆ PAUSE AND DECIDE

What is the most likely cause?

A The environment defined in the bundle does not include a transitive dependency the notebook panel resolved automatically.

Both specify the same library and only one works, so the difference is in what else got resolved alongside it.

B Wheel tasks cannot use environments; only notebook tasks can.

The reverse. Notebook tasks are the ones that cannot take a task environment specification.

C Serverless does not support third-party libraries.

It supports them through environments, which is what the bundle is using.

D The bundle must install libraries at cluster scope rather than in an environment.

There is no cluster to scope to on serverless.

ON THE EXAM

B is the inversion trap. The exception genuinely exists, and it applies to notebook tasks, not wheel tasks.

> A team on the Core pipeline edition must build an SCD Type 2 dimension. Their source is a legacy system with no change data capture, delivering a nightly full extract to cloud storage.
>
> They plan to write the pipeline in SQL, as nobody on the team writes Python.

What must change?

A. Nothing; AUTO CDC FROM SNAPSHOT works in SQL on any edition.

B. Move to serverless or Pro or Advanced, and accept that snapshot processing is Python only.

C. Enable change data capture on the legacy source, which AUTO CDC requires.

D. Upgrade the Databricks Runtime, which unlocks the CDC APIs.

◆ PAUSE AND DECIDE

What must change?

A Nothing; AUTO CDC FROM SNAPSHOT works in SQL on any edition.

Wrong twice. The edition gate is real, and snapshot processing has no SQL interface.

B Move to serverless or Pro or Advanced, and accept that snapshot processing is Python only.

Two independent blockers, and both must be cleared. Their SQL preference cannot be met.

C Enable change data capture on the legacy source, which AUTO CDC requires.

Would let them use the SQL interface, and the snapshot API exists precisely so this is not required.

D Upgrade the Databricks Runtime, which unlocks the CDC APIs.

The gate is the pipeline edition, not the runtime version.

ON THE EXAM

Two blockers, not one. Scan for every stated constraint before choosing, because these questions often hide a second.

> A job's ingestion task loads files and its three downstream transformation tasks are expensive. On days when no files arrive, ingestion succeeds having loaded nothing, and the transformations run anyway.
>
> The team wants the transformations skipped on those days.

What should they build?

A Set the transformations' run-if condition to At least one succeeded.

B Have ingestion set a task value with the row count, then branch on it with an If/else task.

C Make the ingestion task fail when no files arrive, so downstream tasks are skipped.

D Set the transformations' run-if condition to None failed.

◆ PAUSE AND DECIDE

What should they build?

A Set the transformations' run-if condition to At least one succeeded.

Run-if branches on outcome. Ingestion succeeds whether or not it found anything.

B Have ingestion set a task value with the row count, then branch on it with an If/else task.

Branching on a value a previous task computed is exactly what task values and If/else exist for.

C Make the ingestion task fail when no files arrive, so downstream tasks are skipped.

Makes a normal day look like an incident, and poisons alerting and run history.

D Set the transformations' run-if condition to None failed.

Also branches on outcome, and nothing failed.

ON THE EXAM

Outcome or value is the first question. Nothing failed here, so no run-if condition can express what is wanted.

A team must load a one-off historical archive of 1.2 million Parquet files from a decommissioned system into a Delta table, then never read that location again. They work in SQL and want the simplest approach.

A colleague insists Auto Loader is always the recommended tool for file ingestion.

What is the better assessment?

A The colleague is right; Auto Loader is always preferred.

B COPY INTO suits a bounded one-off load, is idempotent on rerun, and leaves no checkpoint to maintain.

C Neither works at this scale; use a Spark batch read and insert.

D Auto Loader is required because COPY INTO does not support Parquet.

◆ PAUSE AND DECIDE

What is the better assessment?

A The colleague is right; Auto Loader is always preferred.

The recommendation is for incremental ingestion of arriving files. This load is bounded and then finished.

B COPY INTO suits a bounded one-off load, is idempotent on rerun, and leaves no checkpoint to maintain.

Idempotence covers the retry risk, and there is no ongoing stream to operate afterwards.

C Neither works at this scale; use a Spark batch read and insert.

Both handle far more. Auto Loader alone scales to billions of files.

D Auto Loader is required because COPY INTO does not support Parquet.

Both support the same eight formats, Parquet included.

ON THE EXAM

Recommendations have scope. Auto Loader is recommended for incremental ingestion, and this load has a defined end.

> A report computes a running total of monthly revenue ordered by month. Several months happen to have identical revenue figures. The engineer wrote an ORDER BY with no explicit frame.
>
> The running total jumps by several months' revenue at once rather than increasing month by month.

What explains it?

A The default frame is RANGE, so rows sharing an ordering value are treated as one peer group.

B The ORDER BY needs a PARTITION BY to reset correctly.

C The default frame includes UNBOUNDED FOLLOWING, so future rows are counted.

D Running totals require a cumulative sum function rather than a window aggregate.

◆ PAUSE AND DECIDE

What explains it?

A The default frame is RANGE, so rows sharing an ordering value are treated as one peer group.

With ORDER BY and no explicit frame the default is RANGE, and peers are included together.

B The ORDER BY needs a PARTITION BY to reset correctly.

A running total across all months is what was wanted.

C The default frame includes UNBOUNDED FOLLOWING, so future rows are counted.

The default stops at the current row, not the end of the partition.

D Running totals require a cumulative sum function rather than a window aggregate.

A window aggregate is exactly how a running total is written.

ON THE EXAM

Note the detail: the duplicates are in the revenue figures, and the ordering is by month. Read carefully which column ties.

A nightly job joining a 6 TB fact table to a currency reference table has run reliably for two years in twelve minutes. This month it began failing with driver out-of-memory errors.

The query carries a BROADCAST hint on the reference table. New currencies and historical rates have been added over time.

What is the cause?

A The fact table has grown beyond what a broadcast join supports.

B The hint is honoured above the broadcast threshold, so the grown reference table is collected to the driver.

C AQE stopped converting the join because statistics are stale.

D Shuffle partitions must be increased for the larger reference table.

◆ PAUSE AND DECIDE

What is the cause?

A The fact table has grown beyond what a broadcast join supports.

The broadcast side is the small one. The large side is never collected.

B The hint is honoured above the broadcast threshold, so the grown reference table is collected to the driver.

A hint overrides the automatic size check, which is exactly what makes it dangerous as data grows.

C AQE stopped converting the join because statistics are stale.

The hint forces the broadcast. AQE is not making this decision.

D Shuffle partitions must be increased for the larger reference table.

A broadcast join does not shuffle the large side.

ON THE EXAM

Two years of reliability then a sudden failure is the signature. Nothing changed except the size of a table somebody once measured.

A team maintains a curated view that joins four tables and masks two columns. They must make it available to a partner organisation that does not use Databricks.

They add the view to a share and create an open sharing recipient. The view never appears for the recipient.

What should they do?

A Views are Databricks-to-Databricks only. Materialise the view as a table and share that.

B Share the view WITH HISTORY, which makes it visible to open sharing recipients.

C Grant the recipient SELECT on the four underlying tables as well.

D The recipient must install a connector before views appear.

◆ PAUSE AND DECIDE

What should they do?

A Views are Databricks-to-Databricks only. Materialise the view as a table and share that.

A view is a query needing a catalog to resolve against, and a non-Databricks recipient has none. Materialising is the intended pattern.

B Share the view WITH HISTORY, which makes it visible to open sharing recipients.

WITH HISTORY concerns cloud token eligibility for tables. Views are not eligible at all.

C Grant the recipient SELECT on the four underlying tables as well.

Defeats the masking the view exists to provide.

D The recipient must install a connector before views appear.

A connector is needed to read anything, and it cannot make an unsupported asset type appear.

ON THE EXAM

Materialising is the intended pattern, not a workaround. The masking is applied when the table is written.

> A dashboard query takes 55 seconds. The query profile shows wall-clock duration of 53 seconds and aggregated task time of 4 seconds. The query is a simple filter and aggregate over a well-clustered managed table.
>
> No operator shows anything unusual.

Where should the engineer look?

A The query logic, since a simple aggregate should not take 53 seconds.

B Compute availability, since aggregated task time far below wall-clock means tasks waited for nodes.

C Data skew, since one task must dominate.

D File layout, since poor pruning would explain the elapsed time.

◆ PAUSE AND DECIDE

Where should the engineer look?

A The query logic, since a simple aggregate should not take 53 seconds.

The work took four seconds. The query is not the problem.

B Compute availability, since aggregated task time far below wall-clock means tasks waited for nodes.

Documented: task time shorter than wall-clock indicates tasks waiting for available nodes.

C Data skew, since one task must dominate.

Skew shows as a long aggregated task time unevenly distributed, not a short one.

D File layout, since poor pruning would explain the elapsed time.

Bad pruning means more data scanned, which raises task time rather than lowering it.

ON THE EXAM

Four seconds of work inside fifty-three seconds of elapsed time. Forty-nine seconds went somewhere that was not computing.

> A support team needs to monitor data quality trends across four pipelines by querying their event logs. The pipelines use default settings and the support team members are not the owners.
>
> An engineer suggests each support member call the event log function with the pipeline identifiers.

Why will that not work?

A The function is owner-only and its views cannot be shared. Publish the event logs, create views, and grant on those.

B The function requires DBR 13.3 LTS on the support team's compute.

C Event logs must be enabled in Advanced settings before any query works.

D Support members must be made pipeline owners in turn.

◆ PAUSE AND DECIDE

Why will that not work?

A The function is owner-only and its views cannot be shared. Publish the event logs, create views, and grant on those.

Two documented facts, plus the recommended route. The function also applies only to legacy publishing mode.

B The function requires DBR 13.3 LTS on the support team's compute.

A real requirement, and not the blocker here.

C Event logs must be enabled in Advanced settings before any query works.

They exist by default as a hidden table. Publishing changes where they live.

D Support members must be made pipeline owners in turn.

Changes who the pipeline runs as, to solve a read-access problem.

ON THE EXAM

Note also that Databricks recommends a view over the event log for security reasons, not merely convenience.

> A team wants clustering keys chosen and adapted automatically as query patterns change. Half their tables are Unity Catalog managed with predictive optimization enabled; the other half are external tables in storage they manage.
>
> They want one consistent approach.

What is possible?

A CLUSTER BY AUTO on all tables, since it is a table-level setting.

B CLUSTER BY AUTO on the managed tables only. External tables need keys declared manually.

C Neither set can use it; automatic clustering applies only at table creation.

D Both, after running OPTIMIZE FULL on the external tables first.

◆ PAUSE AND DECIDE

What is possible?

A CLUSTER BY AUTO on all tables, since it is a table-level setting.

It requires a Unity Catalog managed table, so external tables are excluded.

B CLUSTER BY AUTO on the managed tables only. External tables need keys declared manually.

Automatic key selection is driven by predictive optimization, which runs only on managed tables.

C Neither set can use it; automatic clustering applies only at table creation.

It can be set on existing unpartitioned tables too.

D Both, after running OPTIMIZE FULL on the external tables first.

OPTIMIZE FULL reclusters data. It does not unlock the automatic form.

ON THE EXAM

The honest answer refuses the premise. They cannot have one consistent approach without converting the external tables.

> A team is sizing compute for two workloads: a Structured Streaming job with modest state, and a nightly OPTIMIZE and VACUUM maintenance job on external tables.
>
> They ask which instance family to choose for each.

What does Databricks recommend?

A Memory optimised for both, since streaming holds state and maintenance rewrites files.

B Compute optimised for both, which the docs name for Structured Streaming and for maintenance jobs.

C Storage optimised for both, since both are IO-heavy.

D General purpose for both, in the absence of specific requirements.

◆ PAUSE AND DECIDE

What does Databricks recommend?

A Memory optimised for both, since streaming holds state and maintenance rewrites files.

Memory optimised is named for machine learning and for heavy shuffle and spill.

B Compute optimised for both, which the docs name for Structured Streaming and for maintenance jobs.

The documented rule of thumb names both workloads for compute optimised explicitly.

C Storage optimised for both, since both are IO-heavy.

Named for workloads benefiting from caching, such as ad-hoc interactive analysis.

D General purpose for both, in the absence of specific requirements.

There are specific requirements here, and the docs name them.

ON THE EXAM

This is the row people get wrong. Streaming and maintenance both want compute optimised, which is not most people's instinct.

> A masked salary table is queried by two teams. One team sees masked values as expected. The other, on an older long-term-support runtime, reports the table appears completely empty with no error message.
>
> Both teams have identical Unity Catalog grants.

What is happening?

A Runtimes below 12.2 LTS do not support column masks and fail securely, returning no data.

B The second team lacks SELECT, which presents as an empty table.

C The masking function returns null on that runtime.

D Column masks require serverless compute.

◆ PAUSE AND DECIDE

What is happening?

A Runtimes below 12.2 LTS do not support column masks and fail securely, returning no data.

Documented: unsupported runtimes fail securely rather than returning unprotected data.

B The second team lacks SELECT, which presents as an empty table.

Missing SELECT produces a permissions error, and the grants are stated to be identical.

C The masking function returns null on that runtime.

That would return rows containing nulls, not an empty table.

D Column masks require serverless compute.

Closer to the ABAC requirement. Table-level masks work on classic compute.

ON THE EXAM

The scenario rules out permissions explicitly. An empty result with no error is the protection working correctly.

> A right-to-erasure request must be propagated to a silver streaming table fed from a bronze Delta table. An engineer deletes the records from bronze only, and the pipeline fails on the next run.
>
> They want the deletion to propagate correctly.

What is the documented method?

A Delete from bronze, then fully refresh the streaming table.

B Delete in the source, delete from the streaming table with DML, then set skipChangeCommits on the read.

C Set skipChangeCommits on the read, which handles the deletion automatically.

D Enable change data feed on bronze, which propagates deletes without further work.

◆ PAUSE AND DECIDE

What is the documented method?

A Delete from bronze, then fully refresh the streaming table.

A documented alternative, and it risks data loss where a source is beyond its retention period.

B Delete in the source, delete from the streaming table with DML, then set skipChangeCommits on the read.

The three documented steps, in order. The flag comes last, after the deletion is applied to both tables.

C Set skipChangeCommits on the read, which handles the deletion automatically.

Skips the change without applying it downstream, so silver keeps the record that was supposed to be erased.

D Enable change data feed on bronze, which propagates deletes without further work.

Streaming from the change feed is the more robust design, and it requires handling deletes in your code.

ON THE EXAM

C is the same flag as B, without the two deletions. Order is what separates the documented method from the failure mode.

> An engineer owns a schema and can create, alter and drop every table in it. Running SHOW GRANTS on one of those tables returns no privileges for them at all.
>
> They raise a ticket asking why their permissions have been removed.

What should they be told?

A Permissions were removed and must be re-granted from the schema down.

B Schema owners get implied management of child objects. Implied capabilities do not appear in SHOW GRANTS.

C SHOW GRANTS only lists grants made by the current user.

D They must be metastore admin to see grants on tables they do not own.

◆ PAUSE AND DECIDE

What should they be told?

A Permissions were removed and must be re-granted from the schema down.

Nothing was removed. They can still administer the tables, which the scenario states.

B Schema owners get implied management of child objects. Implied capabilities do not appear in SHOW GRANTS.

Ownership and its implied capabilities are not explicitly granted, so a grants listing shows nothing.

C SHOW GRANTS only lists grants made by the current user.

Invented.

D They must be metastore admin to see grants on tables they do not own.

They can view grants. The point is that their own capability is implied rather than listed.

ON THE EXAM

An empty grants list on an object you can administer is expected. It catches people auditing permissions.

> A cluster starts and reports healthy. Every notebook attached to it fails within seconds with import errors for a library the team installs through an init script. The driver logs show only those import failures.
>
> Log delivery is configured to a storage location.

Where should they look?

A The executor logs, since library installation happens on the workers.

B The init script logs in the delivered log location, since a failed init script leaves a cluster that starts but cannot run anything.

C The compute metrics UI, to check whether memory prevented installation.

D The cluster event log, which records library installation.

◆ PAUSE AND DECIDE

Where should they look?

A The executor logs, since library installation happens on the workers.

Would show the same downstream symptom, not the cause.

B The init script logs in the delivered log location, since a failed init script leaves a cluster that starts but cannot run anything.

The signature failure mode, and log delivery is configured so the init script folder is available.

C The compute metrics UI, to check whether memory prevented installation.

Metrics show resource usage over time, not setup failures.

D The cluster event log, which records library installation.

It records lifecycle events such as creation, termination and resizing.

ON THE EXAM

The import errors are the symptom, faithfully reported. The cause is one layer earlier, at cluster startup.

> A nightly job failed because a source file arrived under the wrong date path. The file is now available under yesterday's date. The team wants to complete that run without altering tomorrow's scheduled behaviour.
>
> A colleague proposes editing the job's default date parameter, repairing, then editing it back.

What is the better approach?

A The colleague's approach is fine if the second edit is not forgotten.

B Repair the run, entering the corrected date as a parameter override, which applies to that repair only.

C Restart the job, which prompts for parameters at launch.

D Create a one-off job with the corrected date.

◆ PAUSE AND DECIDE

What is the better approach?

A The colleague's approach is fine if the second edit is not forgotten.

Works, and mutates a production job to fix one run, with a manual step nobody will remember at 3am.

B Repair the run, entering the corrected date as a parameter override, which applies to that repair only.

Overrides are scoped to the repair, leaving the job definition and tomorrow's schedule untouched.

C Restart the job, which prompts for parameters at launch.

Repeats every task when only the failed one needs running.

D Create a one-off job with the corrected date.

Leaves the original run failed in the history, and adds a second job.

ON THE EXAM

On a later repair, clearing the key and value returns a parameter to its original value, which is a nice touch worth knowing.

> A team version-controls every notebook in a production Git folder updated automatically on merge. Job schedules, cluster configuration and permissions are set by hand in the workspace.
>
> After an outage, they cannot determine what a job's schedule was three weeks ago.

What is missing?

A Enable audit logging to reconstruct configuration history.

B Git folders deploy code, not resources. Defining jobs in a bundle puts schedules, compute and permissions under version control.

C Move the production Git folder under a user folder so history is retained.

D Job configuration cannot be version controlled on Databricks.

◆ PAUSE AND DECIDE

What is missing?

A Enable audit logging to reconstruct configuration history.

Audit records who changed something. It is not the configuration in source control, nor is it redeployable.

B Git folders deploy code, not resources. Defining jobs in a bundle puts schedules, compute and permissions under version control.

Half the deployment is versioned and half is not, and the half that is not is where the question came from.

C Move the production Git folder under a user folder so history is retained.

User-level folders are development checkouts, and location is not the problem.

D Job configuration cannot be version controlled on Databricks.

That is exactly what bundles are for.

ON THE EXAM

The documentation's own division: Git folders for interactive development, bundles for CI/CD and production deployment.

> A sensor readings table has columns for device id, reading value, latitude, longitude, a captured-at timestamp, and a reading date. It receives around 80 million readings per day from 12,000 devices.
>
> Analysts almost always filter by day, occasionally by device.

Which column should be used for partitioning?

A device_id

B captured_at

C reading_date

D latitude

◆ PAUSE AND DECIDE

Which column should be used for partitioning?

A **device_id**

Twelve thousand devices means twelve thousand partitions, and unevenly sized ones.

B **captured_at**

A timestamp at 80 million readings a day is effectively unique. The classic trap.

C **reading_date**

Low cardinality, evenly distributed, useful partition size, and exactly what analysts filter on.

D **latitude**

A float with no relationship to how the data is queried.

ON THE EXAM

The same shape as the exam guide's own published question. Low cardinality, queries filter on it, evenly distributed, useful size.

> A team is building a customer dimension in a new gold schema. Surrogate keys will be generated by an identity column. No keys are being migrated from an existing system.
>
> They ask which identity column form to use.

Which is appropriate, and why?

A Generated by default, which allows values to be supplied if ever needed.

B Generated always, which prohibits supplying your own values and is safer for a new surrogate.

C Either; the two forms are equivalent in behaviour.

D Neither; surrogate keys should be hashes of the business key.

◆ PAUSE AND DECIDE

Which is appropriate, and why?

A Generated by default, which allows values to be supplied if ever needed.

Permitting supplied values on a brand new surrogate invites a conflict with the generated sequence.

B Generated always, which prohibits supplying your own values and is safer for a new surrogate.

Nothing is being migrated, so nothing needs to supply its own values. Prohibiting them removes a class of error.

C Either; the two forms are equivalent in behaviour.

They differ precisely on whether your own values may be supplied.

D Neither; surrogate keys should be hashes of the business key.

A hash is the documented alternative when identity columns are unavailable, such as on an AUTO CDC target.

ON THE EXAM

D is worth remembering for the case it does fit: identity columns are unsupported on AUTO CDC targets.

How to read your result

- **17 or more across all three exams.** You are ready. Book it.
- **13 to 16 consistently.** Close. Identify the two weakest sections and re-watch those modules before booking.
- **Below 13.** Work the misses from all three exams, then re-attempt them in a fortnight.
- Look at the **trend across the three**, not any single score. Improving is a better signal than a high one-off.

> ON THE EXAM
>
> Sixty questions is a reasonable sample. If one section has gone badly in all three, that is a real signal rather than bad luck.

Where this exam appears to be heading

- **Declarative over hand-written.** Pipelines are the recommended default, and hand-written streaming is the exception you justify.
- **Managed over external.** Managed tables are the recommended type, and most automatic optimisation only reaches them.
- **Automatic over configured.** Predictive optimization, automatic clustering, file events, enhanced autoscaling.
- **Policy over per-object.** ABAC over per-table masks, governed tags over conventions, bundles over hand configuration.

> ON THE EXAM
>
> Four movements in the same direction: declare the intent and let the platform decide the mechanics. Questions increasingly reward that instinct.

What that means for answering questions

- When two options are both workable, the one that **declares intent** rather than configuring mechanics is usually intended.
- But **not always.** Module 10 chose COPY INTO over Auto Loader, and module 22 defended partitioning for metadata queries.
- The exam still rewards **reading the requirement**, and a requirement can rule out the modern answer.
- And the guide lags the platform. **Answer the guide's language**, and know what the docs now say.

> ON THE EXAM
>
> A heuristic, not a rule. The trend tells you where to look first; the requirement sentence still decides.

That is the course

- Thirty-eight modules. Ten sections. Sixty practice questions and over two hundred worked scenarios.
- Every fact checked against current documentation, with the places the guide has fallen behind flagged as they arose.
- **Check the exam guide two weeks before you sit.** It changes, and it says so itself.
- Good luck.

> ON THE EXAM
>
> If something in this course turns out to be wrong or out of date, that is worth knowing. The platform moves faster than any course can.